PART 1: Measure-Theoretic Foundations

Mathematical reading notes — revised edition, 2026

Topics. Motivations for the modern definition of probability; sigma-algebras and measurable sets (§1.2); measures and the Dirac measure (§1.3); monotonicity, continuity of measures and the union bound (§1.4). The course's treatment of discrete probability mass functions (§1.6) is not repeated here.

1. Why the classical descriptions of probability are not general definitions

Classical probability

For a finite set S whose outcomes are *equally likely*, the formula

$$P(A) = \frac{|A|}{|S|}$$

provides a useful model. However, it does not itself describe infinite sample spaces, and equal likelihood cannot simply be assumed for arbitrary experiments. In particular, a uniform probability distribution on *all* positive integers does not exist: if every singleton had the same mass c, countable additivity would give $1 = \sum_{n\geq 1} c$, which is impossible for $c \geq 0$.

Relative frequencies

A familiar interpretation of probability is a limiting relative frequency,

$$P(A) \approx \lim_{n\to\infty} \frac{\text{number of occurrences of } A \text{ in the first } n \text{ trials}}{n},$$

when such a limit exists. This is an important consequence or interpretation of a probabilistic model, but it cannot replace the axioms in general: limits of observed frequencies may fail to exist, and functions obtained as limiting *densities* on subsets of $\mathbb{N}$ need not be countably additive.

For example, the asymptotic density $d(A) = \lim_{n\to\infty} |A \cap \{1, \ldots, n\}|/n$, when it exists, gives $d(\{k\}) = 0$ for each singleton $\{k\}$, but $d(\mathbb{N}) = 1$. If it were a probability measure on all subsets, countable additivity would imply $1 = d(\mathbb{N}) = \sum_{k\geq 1} d(\{k\}) = 0$. This is an example of the *failure of countable additivity for asymptotic density*, not a genuine uniform random choice from $\mathbb{N}$.

Geometric probability and a nonmeasurable set

For some finite intervals or regions, probability can be modeled by a ratio of lengths, areas or volumes. Such a model presupposes that the sets under discussion have a well-defined measure. In general one cannot consistently assign an extension of ordinary length to *all* subsets of $\mathbb{R}$ while preserving translation invariance and countable additivity. The following Vitali-set construction illustrates why.

Define an equivalence relation on $[0, 1]$ by

$$x \sim y \quad \Longleftrightarrow \quad x - y \in \mathbb{Q}.$$

The interval is partitioned into pairwise disjoint equivalence classes $[x] = \{y \in [0, 1] : y - x \in \mathbb{Q}\}$. Each class is countable, because its elements form a subset of a rational translate; there are uncountably many classes, since a countable union of countable sets would be countable. By the axiom of choice, select one point from each class, and let $V \subset [0, 1]$ be the set of representatives (a Vitali set).

Distinct points of V cannot differ by a rational number. Consider *rational* translates of V, indexed by $q \in \mathbb{Q}\cap[-1, 1]$:

$$V + q = \{v + q : v \in V\}, \qquad A = \bigcup_{q\in\mathbb{Q}\cap[-1,1]} (V + q).$$

These translates are pairwise disjoint: if $v_1 + q_1 = v_2 + q_2$, then $v_1 - v_2 = q_2 - q_1 \in \mathbb{Q}$, so by construction $v_1 = v_2$ and hence $q_1 = q_2$.

Every $x \in [0, 1]$ differs by some rational $q \in [-1, 1]$ from the representative of its class; therefore $x \in V + q$. Conversely $V \subset [0, 1]$ and $q \in [-1, 1]$, so each translate is contained in $[-1, 2]$. Thus

$$[0, 1] \subseteq A \subseteq [-1, 2].$$

Proposition 1 (Vitali contradiction)**.** *There is no countably additive, translation-invariant, nonnegative measure m defined on* every subset *of $\mathbb{R}$ that assigns ordinary length to every interval.*

Proof. If such a measure existed, monotonicity and the displayed inclusions would imply $1 \le m(A) \le 3$. On the other hand, using disjointness, translation invariance and countable additivity,

$$m(A) = \sum_{q \in \mathbb{Q} \cap [-1,1]} m(V+q) = \sum_{q \in \mathbb{Q} \cap [-1,1]} m(V).$$

There are infinitely many such rationals. If $m(V) = 0$ the sum is 0; if $m(V) > 0$ (including $+\infty$) the sum is $+\infty$. Both contradict $1 \le m(A) \le 3$. □

The argument *does not* say that points of A lying outside $[0,1]$ have measure zero, nor that A has the same length as $[0,1]$. It motivates restricting the domain of measurable events instead of attempting to measure every subset.

Further details of the Vitali construction

The equivalence classes in the preceding argument are neither singletons nor a finite partition. For example, if $x \in [0,1]$, then every $x+q$ that stays inside $[0,1]$, with $q \in \mathbb{Q}$, belongs to the class of x. Since $\mathbb{Q}$ is countable, so is each equivalence class. The axiom of choice is used to select one representative *simultaneously* from each class. It does not prescribe an effective procedure for listing those representatives.

The restriction to *rational* translates is crucial to disjointness. If q_1, q_2 were arbitrary distinct *real* numbers, then $q_2 - q_1$ need not be rational, and the defining property of a Vitali set would not yield a contradiction. The corrected measure argument uses both inclusions

$$[0,1] \subseteq \bigcup_{q \in \mathbb{Q} \cap [-1,1]} (V+q) \subseteq [-1,2],$$

not an equality of the middle set with $[0,1]$. This is why monotonicity gives lower and upper *bounds* on its length.

A final point: this construction disproves the existence of a length assignment to *every* real subset satisfying the stated properties. It does not say that geometric probability is invalid on its natural domain. Lebesgue measure is well defined on the sigma-algebra of Lebesgue-measurable sets, so ordinary geometric probability remains a valuable special case.

2. Probability spaces and sigma-algebras (§1.2)

A *sigma-algebra* $\mathcal{S}$ on a set S is a collection of subsets satisfying

(i) $\varnothing, S \in \mathcal{S}$;

(ii) $A \in \mathcal{S} \implies S \setminus A \in \mathcal{S}$;

(iii) $A_1, A_2, \ldots \in \mathcal{S} \implies \bigcup_{n \ge 1} A_n \in \mathcal{S}$.

By De Morgan's law it is also closed under countable intersections. Members of $\mathcal{S}$ are *measurable sets*; $(S, \mathcal{S})$ is a measurable space. Both the full power set $\mathcal{P}(S)$ and the trivial sigma-algebra $\{\varnothing, S\}$ are examples, for *any* set S.

A *probability space* is a triple $(S, \mathcal{S}, P)$, where $P : \mathcal{S} \to [0,1]$ is a probability measure. We do not ask P to assign probabilities to all subsets of S unless we have specifically chosen the full power set as the sigma-algebra.

How the axioms are related

The requirement that $\mathcal{S}$ be closed under countable complements and unions is exactly what is needed when defining events such as "one of the A_n happens", "none of the A_n happens", and "all A_n happen". For example, countable intersections follow without a separate axiom:

$$\bigcap_{n \ge 1} A_n = S \setminus \bigcup_{n \ge 1} (S \setminus A_n).$$

Finite unions and intersections follow as special cases. In a finite sample space the power set is a natural choice, whereas for $S = \mathbb{R}$ the Borel or completed Lebesgue sigma-algebras are standard choices. Thus a probability space is a *sample space plus an explicit collection of admissible events and their probabilities*, not just a sample space with an informal frequency interpretation.

3. Measures and probability measures (§1.3)

A measure on $(S, \mathcal{S})$ is a map $\mu : \mathcal{S} \to [0, \infty]$ such that

$$\mu(\varnothing) = 0, \qquad \mu\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \mu(A_n)$$

whenever the measurable sets A_n are pairwise disjoint. The second property is *countable additivity*. A finite measure satisfies $\mu(S) < \infty$; a probability measure satisfies $\mu(S) = 1$. If $0 < \mu(S) < \infty$, normalizing μ by $\mu(S)$ produces a probability measure.

The Dirac measure

For a fixed point $x \in S$, define

$$\delta_x(A) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A, \end{cases} \qquad A \in \mathcal{S}.$$

It is a probability measure: clearly $\delta_x(\varnothing) = 0$ and $\delta_x(S) = 1$. For pairwise disjoint A_n, either x belongs to none of them, in which case both sides of countable additivity equal zero, or x belongs to exactly one A_k, in which case both sides equal one.

Countable additivity illustrated

Take mutually disjoint measurable sets $A_1, A_2, \ldots$. For a probability measure P, the identity

$$P\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} P(A_n)$$

means that the probability of one of the disjoint alternatives is obtained by adding their individual probabilities, even for infinitely many alternatives. For example, when $A_n = \{n\}$ in the space of positive integers, the point masses must sum to one. An attempt to assign the same strictly positive probability to all positive integers would make the sum infinite; assigning zero to all of them would make it zero. This recovers the obstruction encountered in the discussion of uniform probability on $\mathbb{N}$.

4. Properties of measures (§1.4)

Monotonicity

If $A \subseteq B$ are measurable, then $B = A \,\dot\cup\, (B \setminus A)$. Consequently

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A).$$

Thus measures are monotone.

Increasing and decreasing sequences of sets

Write $A_n \uparrow A$ when $A_1 \subseteq A_2 \subseteq \cdots$ and $\bigcup_{n\geq 1} A_n = A$. Write $A_n \downarrow A$ when $A_1 \supseteq A_2 \supseteq \cdots$ and $\bigcap_{n\geq 1} A_n = A$. For example, for $n \geq 1$,

$$[0, 1 - 1/n] \uparrow [0, 1), \qquad [-1/n, 1] \downarrow [0, 1].$$

In the first example the right endpoint 1 is *not* part of the union; that matters for the set-theoretic statement even though the Lebesgue measures of $[0, 1)$ and $[0, 1]$ coincide.

Proposition 2 (Continuity of measures)**.** *For measurable sets,*

$$\begin{aligned} A_n \uparrow A &\implies \mu(A_n) \uparrow \mu(A), \\ A_n \downarrow A \text{ and } \mu(A_1) < \infty &\implies \mu(A_n) \downarrow \mu(A). \end{aligned}$$

Proof. For increasing sets, let $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. These sets are disjoint and satisfy $A_n = \dot{\bigcup}_{j=1}^{n} B_j$ and $A = \dot{\bigcup}_{j\geq 1} B_j$. Countable additivity gives

$$\lim_{n\to\infty} \mu(A_n) = \sum_{j\geq 1} \mu(B_j) = \mu(A).$$

For decreasing sets with $\mu(A_1) < \infty$, set $B_n = A_1 \setminus A_n$ and $B = A_1 \setminus A$. Then $B_n \uparrow B$. The first part gives $\mu(B_n) \uparrow \mu(B)$, and

$$\mu(A_n) = \mu(A_1) - \mu(B_n) \longrightarrow \mu(A_1) - \mu(B) = \mu(A).$$

The finiteness assumption permits these subtractions. □

This result lets us compute the probability of limiting events by approximating them with simpler increasing or decreasing sequences of events.

Examples and the role of finite measure

For Lebesgue measure λ, let $A_n = [0, 1 - 1/n]$ for $n \geq 1$. These sets increase to $[0, 1)$, and $\lambda(A_n) = 1 - 1/n \uparrow 1 = \lambda([0, 1))$. For the decreasing sets $B_n = [-1/n, 1]$, the intersection is $[0, 1]$, and $\lambda(B_n) = 1 + 1/n \downarrow 1$.

The finiteness assumption in continuity from above is not merely a technicality. Consider $C_n = [n, \infty)$ under Lebesgue measure. Then $C_n \downarrow \varnothing$, but $\lambda(C_n) = \infty$ for every n; the measures do not decrease to the measure 0 of the empty intersection. This example explains why the subtraction $\mu(A_1) - \mu(B_n)$ in the proof is legitimate only when $\mu(A_1) < \infty$. For probability measures the condition always holds because $P(S) = 1$.

The usefulness of continuity from below can be expressed in terms of limiting events. If A_n represents "an event has happened by time n", then the events A_n are typically increasing. The probability of "it happens at some finite time" is then the limit of $P(A_n)$. Likewise, continuity from above handles progressively restrictive events when the initial probability is finite.

The union bound

For any finite or countable family of measurable sets (A_i),

$$\mu\left(\bigcup_{i\geq 1} A_i\right) \leq \sum_{i\geq 1} \mu(A_i).$$

For two sets A, B, since $A \cup B = A \,\dot\cup\, (B \setminus A)$,

$$\mu(A \cup B) = \mu(A) + \mu(B \setminus A) \leq \mu(A) + \mu(B).$$

Induction gives the finite version. For countably many sets, set $C_n = \bigcup_{i=1}^{n} A_i$, note $C_n \uparrow \bigcup_{i\geq 1} A_i$, and use continuity from below:

$$\mu\left(\bigcup_{i\geq 1} A_i\right) = \lim_{n\to\infty} \mu(C_n) \leq \lim_{n\to\infty} \sum_{i=1}^{n} \mu(A_i) = \sum_{i\geq 1} \mu(A_i).$$

For disjoint sets equality holds by countable additivity; equality may also hold for overlapping sets when their overlaps have measure zero.

An alternative disjointification proof

The union bound does not require its events to be disjoint. For a sequence $A_1, A_2, \ldots$, define

$$B_1 = A_1, \qquad B_n = A_n \setminus \bigcup_{j=1}^{n-1} A_j \quad (n \geq 2).$$

Then the B_n are pairwise disjoint, $B_n \subseteq A_n$, and $\bigcup B_n = \bigcup A_n$. By countable additivity and monotonicity,

$$\mu\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \mu(B_n) \leq \sum_{n\geq 1} \mu(A_n).$$

This proof shows directly why the union bound is so useful: it converts overlapping alternatives into disjoint ones and then bounds each newly created disjoint piece by its original event. The same idea will reappear when comparing probability laws through overlapping mass.

Reading note

This part introduces the measure-theoretic language needed to define distributions and couplings in the subsequent parts. The Vitali construction is presented to explain why one works with a sigma-algebra of measurable events rather than arbitrary subsets.